INCH-POUND

MS51832E
9 May 2022
SUPERSEDING
MS51832D
w/AMENDMENT 1
20 November 2012

DETAIL SPECIFICATION SHEET

# INSERT, SCREW – THREAD, LOCKED IN, KEY-LOCKED, EXTRA HEAVY DUTY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification MIL-I-45914. In the event of a conflict between this specification sheet and MIL-I-45914, this specification sheet shall take precedence.

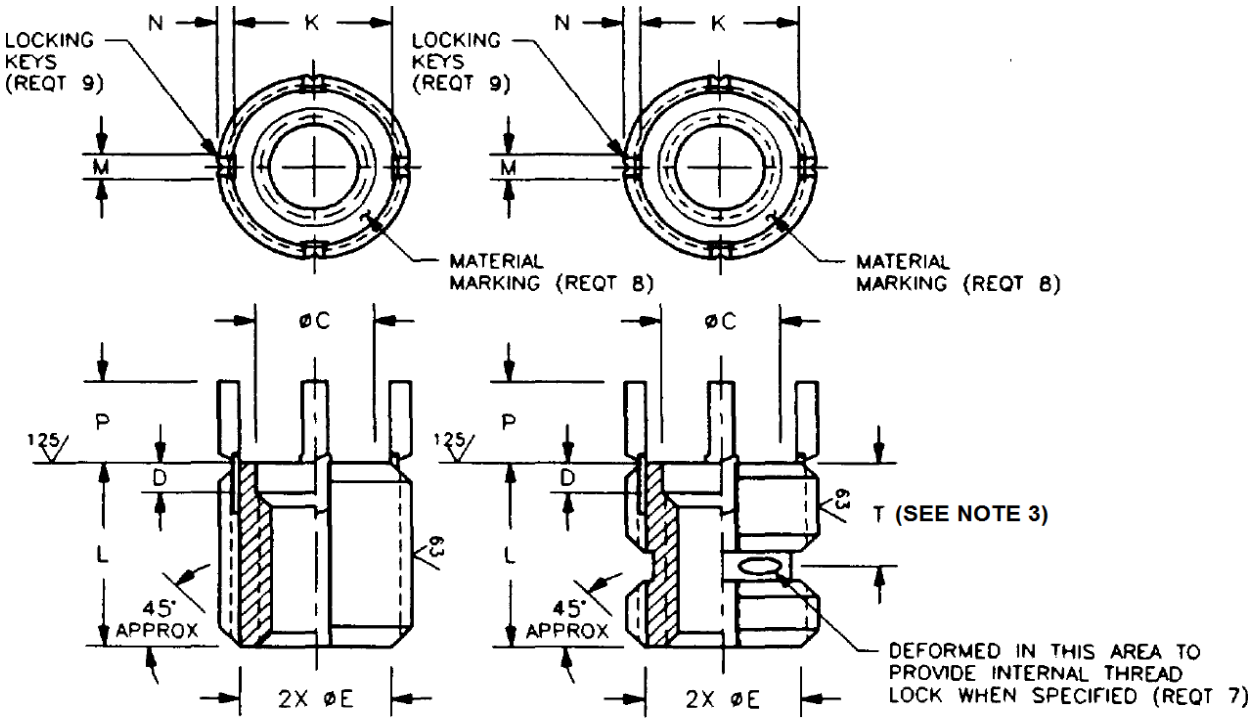



FIGURE 1. Insert, coarse internal threads.

Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency - Aviation, ATTN: DLA Aviation-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.


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TABLE I. Coarse internal threads.

| DASH NO. | INTERNAL THREAD SAE AS8879 UNJC-3B | EXTERNAL THREAD FED-STD-H28/2 EXCEPT MODIFIED MINOR DIA. | | ØC ±.005 | D ±.010 | ØE REF | K REF | L ±.015 | KEY DIMENSIONS | | | T ±.030 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SIZE | MINOR DIA | | | | | | M REF | N REF | P REF | |
| 101 | .190-24 | .4375-14UNC-2A | .383<br>.375 | .196 | .070 | .375 | .344 | .312 | .068 | .040 | .130 | .220 |
| 102 | .250-20 | .500-13UNC-2A | .440<br>.432 | .257 | .070 | .430 | .407 | .375 | .068 | .040 | .190 | .220 |
| 103 | .3125-18 | .5625-12UNC-2A | .503<br>.495 | .316 | .070 | .495 | .469 | .437 | .068 | .040 | .190 | .220 |
| 104 | .375-16 | .625-11UNC-2A | .565<br>.551 | .380 | .070 | .550 | .532 | .500 | .068 | .040 | .190 | .250 |
| 105 | .4375-14 | .6875-11NS-2A | .625<br>.615 | .445 | .100 | .615 | .595 | .625 | .068 | .040 | .190 | .280 |
| 106 | .500-13 | .8125-16UN-2A | .752<br>.744 | .507 | .100 | .745 | .720 | .688 | .068 | .040 | .190 | .280 |
| 107 | .5625-12 | .875-14UNF-2A | .815<br>.807 | .580 | .100 | .805 | .782 | .812 | .068 | .040 | .190 | .320 |
| 108 | .625-11 | 1.000-12UNF-2A | .924<br>.916 | .640 | .100 | .915 | .850 | .875 | .103 | .070 | .250 | .330 |
| 109 | .750-10 | 1.250-12UNF-2A | 1.175<br>1.165 | .763 | .100 | 1.165 | 1.100 | 1.125 | .103 | .070 | .250 | --- |
| 109L | .750-10 | 1.250-12UNF-2A | 1.175<br>1.165 | .763 | .100 | 1.165 | 1.100 | 1.250 | .103 | .070 | .250 | .490 |
| 110 | .875-9 | 1.375-12UNF-2A | 1.300<br>1.290 | .888 | .100 | 1.290 | 1.225 | 1.250 | .103 | .070 | .250 | --- |
| 110L | .875-9 | 1.375-12 UNF-2A | 1.300<br>1.290 | .888 | .100 | 1.290 | 1.225 | 1.375 | .103 | .070 | .250 | .490 |
| 111 | 1.000-8 | 1.500-12UNF-2A | 1.424<br>1.415 | 1.015 | .100 | 1.415 | 1.350 | 1.375 | .103 | .070 | .250 | --- |
| 111L | 1.000-8 | 1.500-12UNF-2A | 1.424<br>1.416 | 1.015 | .100 | 1.415 | 1.350 | 1.500 | .103 | .070 | .250 | .550 |



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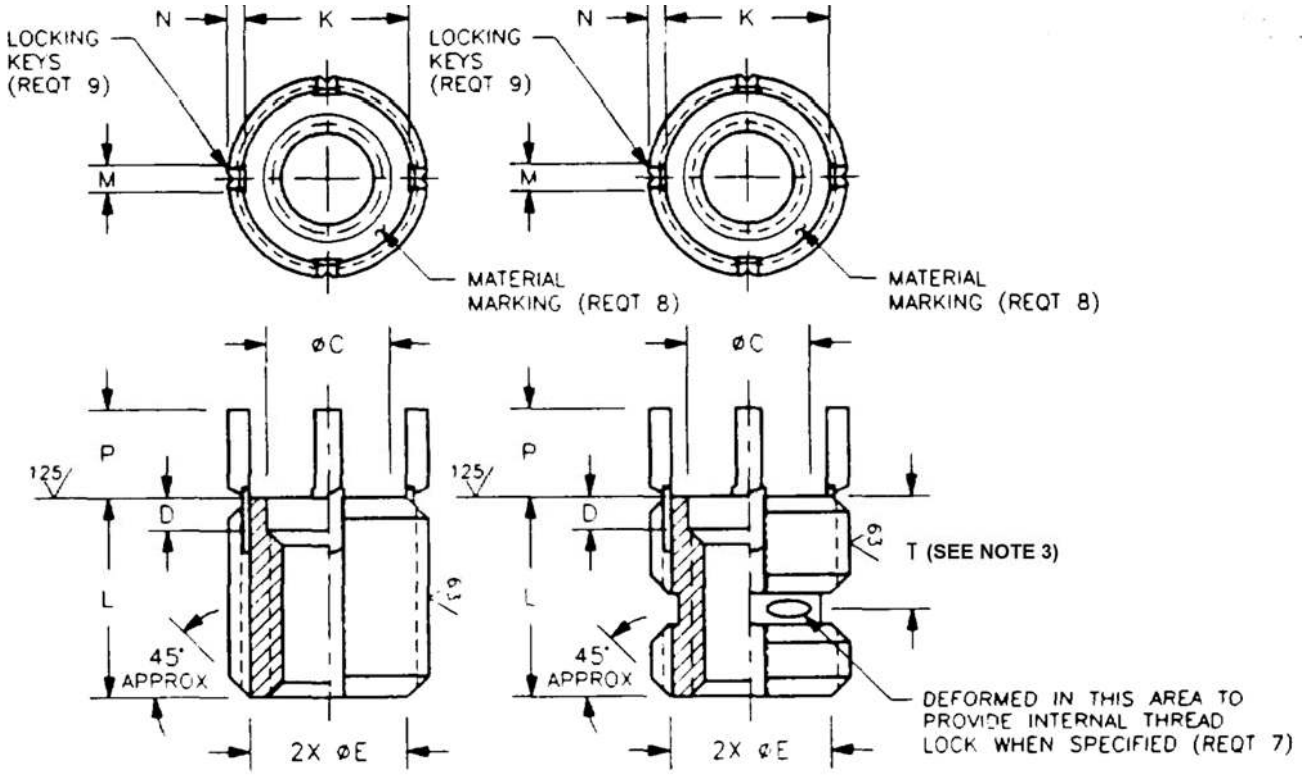



FIGURE 2. Insert, fine internal threads.

TABLE II. Fine internal threads.

| DASH NO. | INTERNAL THREAD SAE AS8879 UNJF-3B | EXTERNAL THREAD FED-STD-H28/2 EXCEPT MODIFIED MINOR DIA. | | ØC ±.005 | D ±.010 | ØE REF | K REF | L ±.015 | KEY DIMENSIONS | | | T ±.030 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SIZE | MINOR DIA | | | | | | M REF | N REF | P REF | |
| 201 | .190-32 | .4375-14UNC-2A | .383<br>.375 | .196 | .070 | .375 | .344 | .312 | .068 | .040 | .130 | .220 |
| 202 | .250-28 | .500-13UNC-2A | .440<br>.432 | .257 | .070 | .430 | .407 | .375 | .068 | .040 | .190 | .220 |
| 203 | .3125-24 | .5625-12UNC-2A | .503<br>.495 | .316 | .070 | .495 | .469 | .437 | .068 | .040 | .190 | .220 |
| 204 | .375-24 | .625-11UNC-2A | .565<br>.551 | .380 | .070 | .550 | .532 | .500 | .068 | .040 | .190 | .250 |
| 205 | .4375-20 | .6875-11NS-2A | .625<br>.615 | .445 | .100 | .615 | .595 | .625 | .068 | .040 | .190 | .280 |
| 206 | .500-20 | .8125-16UN-2A | .752<br>.744 | .507 | .100 | .745 | .720 | .688 | .068 | .040 | .190 | .280 |
| 207 | .5625-18 | .875-14UNF-2A | .815<br>.807 | .580 | .100 | .805 | .782 | .812 | .068 | .040 | .190 | .320 |
| 208 | .625-18 | 1.000-12UNF-2A | .924<br>.916 | .640 | .100 | .915 | .850 | .875 | .103 | .070 | .250 | .330 |
| 209 | .750-16 | 1.250-12UNF-2A | 1.175<br>1.165 | .763 | .100 | 1.165 | 1.100 | 1.125 | .103 | .070 | .250 | --- |
| 209L | .750-16 | 1.250-12UNF-2A | 1.175<br>1.165 | .763 | .100 | 1.165 | 1.100 | 1.250 | .103 | .070 | .250 | .490 |
| 210 | .875-14 | 1.375-12UNF-2A | 1.300<br>1.290 | .888 | .100 | 1.290 | 1.225 | 1.250 | .103 | .070 | .250 | --- |
| 210L | .875-14 | 1.375-12UNF-2A | 1.300<br>1.290 | .888 | .100 | 1.290 | 1.225 | 1.375 | .103 | .070 | .250 | .490 |
| 211 | 1.000-12 | 1.500-12UNF-2A | 1.424<br>1.416 | 1.015 | .100 | 1.415 | 1.350 | 1.375 | .103 | .070 | .250 | --- |
| 211L | 1.000-12 | 1.500-12UNF-2A | 1.424<br>1.416 | 1.015 | .100 | 1.415 | 1.350 | 1.500 | .103 | .070 | .250 | .550 |



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Requirements:

1. Material. Steel, corrosion-resistant, Type 303 (UNS S30300) in accordance with chemical composition of SAE AMS 5640 (Type 1) or ASTM A582/A582M, or Type 303SE (UNS S30323) in accordance with chemical composition of SAE AMS 5640 (Type 2), SAE AMS 5738 or ASTM A582/A582M.

   Steel, corrosion-resistant. Type A286 (UNS S66286) in accordance with SAE AMS 5731, SAE AMS 5734 or SAE AMS 5737.

   Steel, alloy, grade 4140 (UNS G41400) in accordance with SAE AMS 6349 or SAE AMS 6382 or grade 8740 (UNS G87400) in accordance with SAE AMS 6322.

   Locking keys: Steel, corrosion-resistant. Type 302 (UNS S30200) chemical composition of ASTM A580/A580M only.

2. Surface treatment.
   - Steel, corrosion resistant shall be passivated in accordance with SAE AMS2700, processor may choose the method from specification recommendations except the use of hexavalent chromium (dichromate) is prohibited.
   - For designator A, alloy steel shall be cadmium plated in accordance with SAE AMS-QQ-P-416, Type II, Class 3. Cadmium is an R1, R2, and D1 hazardous substance in the Aerospace and Defense Declarable Substance List. DO NOT USE DESIGNATOR “A” FOR NEW DESIGNS.
   - For designator QX, alloy steel shall be zinc-nickel plated per MIL-PRF-32647, Type K16.
   - The locking keys may or may not be plated at manufacturer option. The use of cadmium or hexavalent chromium in the plating system for the locking keys is prohibited.

3. Lubrication. Inserts with self-locking internal threads for aerospace applications shall be dry film lubricated in accordance with SAE AS5272 Type I (See Note 5). Inserts with self-locking internal threads for non-aerospace applications shall be dry film lubricated in accordance with MIL-PRF-46010 (See Note 6). The locking keys may or may not be lubricated.

4. Surface texture: Machined surfaces shall be in accordance with ASME B46.1.

5. Heat treatment: Alloy steel inserts shall be heat treated to 160,000 psi Ftu minimum in accordance with SAE AMS-H-6875. Corrosion-resistant steel inserts, type A286 (SAE AMS 5734) shall be heat treated to 140,000 psi Ftu minimum.

6. Hardness. Alloy steel inserts shall have a hardness range of 36-40 HRC.

7. Part number. The part number shall consist of the basic MS sheet number plus the dash number taken from table I or table II, as applicable:

   Example: MS51832 - 102 L

   - L: Add "L" as suffix to dash number for internal thread lock for aerospace applications (See Req. 3). Add “LM” as suffix to dash number for internal thread lock for non-aerospace applications (See Req. 3). Leave blank if internal thread lock is not required.
   - 102: Dash number from Table I or Table II.
   - -: Material: Dash indicates CRES. Type 303 or 303SE. Add “CA" in lieu of dash for CRES, Type A286. Add “A" in lieu of dash for alloy steel, grade 4140 or 8740, cadmium plated (don’t use for new designs). Add “QX” in lieu of dash for alloy steel, grade 4140 or 8740, zinc-nickel plated.
   - MS51832: Basic MS sheet number.

   Example: MS51832CA102L indicates - Insert, Screw-Thread, Locked In, Key-Locked, Lightweight, CRES A286, .250-20UNJC-3B internal thread with internal thread lock feature.

8. Material marking: CRES 303 or 303SE has no identifying mark.
   CRES A286 shall be identified on top of insert by one (1) line or dash mark.
   Alloy steel 4140 or 8740 shall be identified on top of insert by two (2) parallel lines or dash marks.
   Material marking of miniature type optional.

9. Inserts with internal thread size .190 and smaller shall be supplied with two (2) locking keys spaced 180° apart. Inserts with internal thread size .250 and greater shall be supplied with four (4) locking keys spaced 90° apart.

10. Inserts shall be free of all hanging burrs and slivers which might become dislodged under usage.

11. Source identification mark: Source identification mark shall be in accordance with MIL-I-45914.

12. All dimensions are after surface treatment and prior to the addition of the lubrication.

13. Fillets are R .015 maximum.



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Notes:

1. All dimensions are in inches.

2. Installation of inserts shall be in accordance with MS51835.

3. Distance to center of internal thread lock.

4. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

5. SAE AS5272 Type I lubricant is technically equivalent to MIL-L-46010 Type I lubricant used in previous revisions.

6. MIL-PRF-46010 lubricant is lead (Pb) free and is not technically equivalent to MIL-L-46010 Type I lubricant used in previous revisions. Use of MIL-PRF-46010 in aerospace applications must first be validated.

7. For ground vehicle applications, designator QX, zinc-nickel electroplating as specified herein may be substituted for designator “A.” This allows for a hazardous material-free plating option.

8. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTEREST

Custodians:
Army – AR
Navy – AS
Air Force – 71
DLA – GS

Preparing activity:
DLA – GS6

(Project 5325-2022-011)

Review activities:
Army – AT, AV, CR, MI
Navy – EC, MC, SH

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